McMahan et al. 2017 — Communication-Efficient Learning from Decentralized Data (FedAvg)

Federated Learning & the FederatedAveraging algorithm | arXiv:1602.05629 | my study repo

One-Liner

Train a shared model on data that never leaves the device: each client runs several epochs of local SGD, the server averages the resulting *models*; this cuts communication rounds 10–100× versus synchronized SGD.

What's Novel

- The *problem* and its framing — "federated optimization" defined by four properties (non-IID, unbalanced, massively distributed, limited communication) — not the averaging algorithm, which predates the paper (McDonald et al. 2010; Povey et al. 2015).
- One dominant knob: spend "essentially free" on-device computation (E local epochs, minibatch B) to buy the scarce resource, communication rounds. Speedups come mostly from more local work, not more clients.
- An evaluation methodology for the regime that violates every assumption of convex distributed optimization ($K \ll n_k$, IID, equal n_k).

Math in 60 Seconds

The objective is a finite sum, decomposable exactly over clients (no IID needed):

$$f(w) = \frac{1}{n}\sum_i f_i(w) = \sum_k \frac{n_k}{n} F_k(w).$$

Differentiating gives $\sum_k \frac{n_k}{n} g_k = \nabla f(w_t)$, so **FedSGD** ($B{=}\infty, E{=}1, C{=}1$) is exactly full-batch GD. Because $\sum_k n_k/n = 1$, averaging one local step equals one global step — **FedAvg** just iterates the local step first. For $E > 1$ the per-round gap from centralized GD is, to leading order, $\Delta = \eta^2\,\mathrm{Cov}_k(H_k, g_k)$: a pure client-*heterogeneity* term (zero when IID). Non-convex averaging only works from a *shared* init (Figure 1: independent inits cross a loss barrier); FedAvg re-broadcasts w_t each round to guarantee it.

What I'd Push Back On

- No theory at all — no convergence result, no rate; the $E > 1$ regime is uncharacterized and can "plateau or diverge." The justification for averaging is one 1-D loss-interpolation plot.
- Evaluation flatters the method: per-x-point learning-rate grid search ($>$ 2000 models), monotonized curves, interpolated round counts, no error bars; the FedSGD baseline (one full-batch step/round) is the weakest possible comparator.
- "Computation is free" is asserted, never measured (up to 1200 local updates/round on a phone); the headline 95× is the *unbalanced* Shakespeare case the authors admit is easier, and the erratum's n_k/m_t average is only exactly unbiased when clients are balanced.

My One Proposed Extension

Cancel the heterogeneity drift directly. Since the $E > 1$ "free lunch" is exactly the covariance $\eta^2\mathrm{Cov}_k(H_k, g_k)$, add SCAFFOLD-style control variates so each local step uses $g_{\text{local}} - c_k + c$, re-centering clients onto the global direction at no extra backward pass (SCAFFOLD "Option II"). My prototype recovers 1.62× fewer rounds on the pathological non-IID partition — operationalizing the paper's own weakest spot, and orthogonal to the aggregation-bias fix, the E-decay schedule, and permutation-aligned averaging.

Full proposal set: `05-improvements.tex` *in the study repo.*

My One Question

For $E > 1$ under genuine non-IID data, what fixed point (if any) does FedAvg converge to, and is there a principled rule for choosing (E, B) — given that Table 2 shows more local work sometimes makes things worse*?*